★ SPECIAL SECTION • CONCLUSION

Week 7 is the professional-engineer plumbing around software - and, in its later half, around machine learning systems specifically. It takes the developer who can write code that works and turns them into the engineer who ships code that keeps working, in a team, in production, and under the scrutiny of a growing user base.\n\nThe five chapters build progressively.

Chapter 1 (Git Fundamentals) covers solo version control done correctly - the three states, `.gitignore` discipline, atomic commits, and Conventional Commits format that turn `git log` into a readable history. **Chapter 2 (Git + GitHub Mastery)** extends into team collaboration - branch strategies, the merge-vs-rebase distinction with its golden rule, the conflict-resolution ritual, `git reflog` as the 90-day safety net, PR etiquette (small, focused, well-described), and branch protection as the technical enforcement of quality standards.

Chapter 3 (Testing and CI) adds the automated safety net - pytest fluency with fixtures, parametrize, marks and mocking (the high-emphasis 3.2), coverage discipline with branch coverage as the meaningful metric, CI/CD literacy separating the three flavors of CD, and hands-on GitHub Actions (the high-emphasis 3.5) to gate merges on green tests. **Chapter 4 (CI/CD Delivery + MLOps Intro)** tightens the loop before code leaves your machine (ruff + black + mypy + pre-commit) and extends the pipeline past CI into build-once-deploy-many CD, then pivots to MLOps with MLflow tracking (the high-emphasis 4.4) as Git-for-experiments and the Model Registry with alias-based promotion decoupling model deploys from code deploys.

Chapter 5 (CI/CD for ML + Consolidation) assembles everything into one working system - DVC for data versioning, contract tests as automated promotion gates, orchestrated training pipelines, a six-milestone hands-on lab, and the rehearsed interview walkthrough that ties the whole story together.\n\nAskMyNotes - the running RAG-based PDF Q&A example - graduates from Chapter 1's first `git commit` to Chapter 5's full three-pipeline system with tracked experiments, registered models, alias-based promotion, contract-gated deploys, and full lineage from any production prediction back to git SHA + DVC hash + hyperparameters. That end-state is what interviewers listen for when probing MLOps maturity, and what you'll want to see on the first day of any ML engineering role you take.\n\nThe high-emphasis subsections - **3.2 pytest advanced, 3.5 GitHub Actions, and 4.4 MLflow tracking** - deserve special attention.

They are the daily-use skills every mid-level engineering interview probes hardest, and they give you the portfolio artifacts (green CI badge, workflow YAML, MLflow UI screenshot) that turn 'I know MLOps' from a claim into visible evidence. Return to those subsections whenever you set up a new project; the muscle memory takes several projects to build, and this Refcard is the reference to return to.\n\nNothing in Week 7 is theoretical.

Every command in every code block was chosen because you will type it (or something very close to it) in your first real engineering role. Every workflow file, every configuration, every promotion pattern is the default at hundreds of small-to-medium teams shipping software and ML products in 2026.

Master this material once; it applies for the rest of your career. Beyond Week 7 lies specialization (LLM serving, feature stores, distributed training, kubernetes-native ML platforms, feature engineering at scale) - but every one of those specializations assumes fluency with the practices captured here. This is the foundation on which everything else is built.